Quartus Tutorial02A: 2-bit Adder Design GLaw

Schematic Design Entry Method with Quartus Graphical Simulation

Notation:

To simplify what one should click to perform a task, the following form of notation will be used:

Menu item → Menu item → … → Menu item

Example:

File → Project Wizard means "on the File menu, click New Project Wizard"

Objective

To learn to use Quartus to develop and simulate a structured 2-bit adder using Schematic Design Entry Method

Development Tools

Quartus 7.2

Hardware

DE2 board, USB Blaster, and USB cable.

References

Altera Quartus Handbooks Volume 1, 2, and 3 available at www.altera.com or at the course website www.csun.edu/~glaw

Launch Quartus II Software

1. Launch the Quartus II software.
2. File → Project Wizard (on the File menu, click New Project Wizard)
3. New Project Wizard: Introduction windows appears. It shows you the five steps you may need to go through in setting up a new project. You have nothing to do in this window, hence click **Next**.
4. New Project Wizard: Directory, Name, Top-Level Entity window appears.

   a. Enter the working project directory in your Z drive, for example, z:\...\Tutorial02A

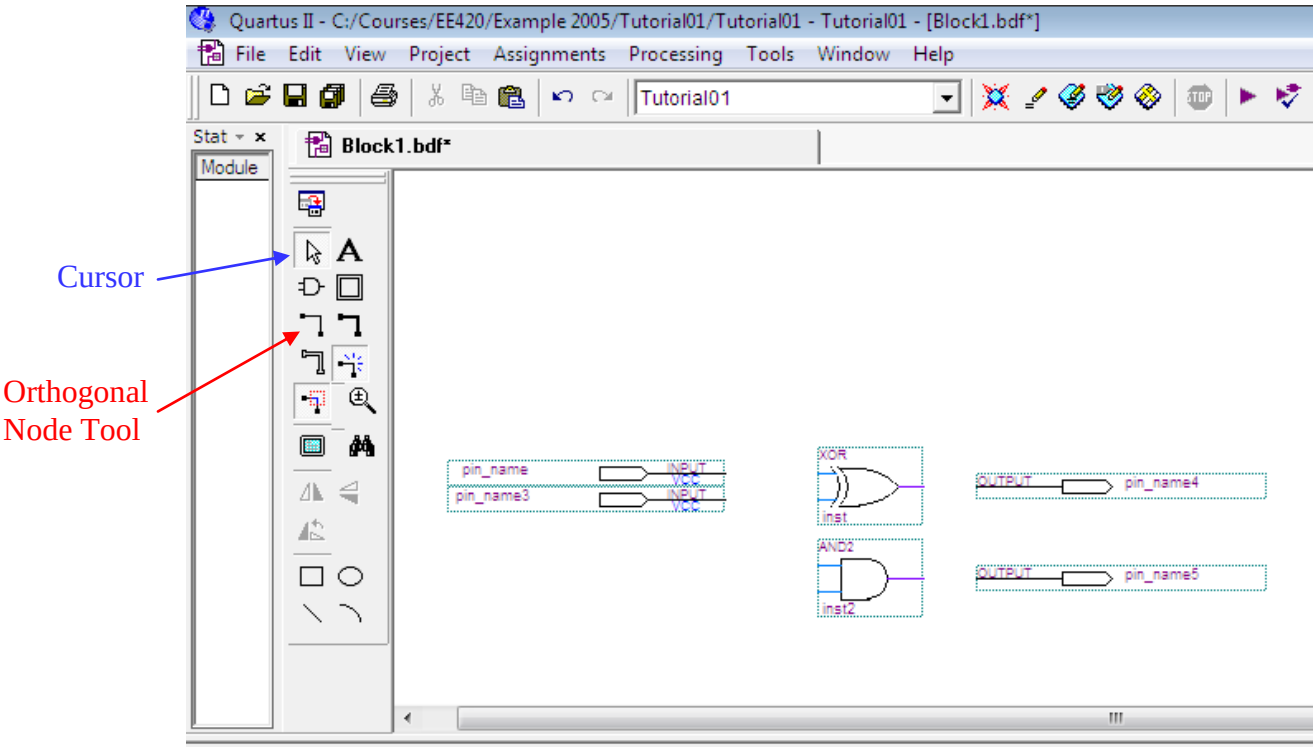
   b. Enter project name: Tutorial02A. As you type Tutorial02A name, Top-Level Entity name will be automatically filled with Tutorial02A.

   c. Click next. When asked to create z:\...\Tutorial02A directory, click **Ye**s.
5. New Project Wizard: Add Files window appears. You have no files to add now, but need to add files later. Hence, just click **Next**.

6. New Project Wizard: Family & Device Settings window appears.
   Scroll up/down the family choices and choose Cyclone II family. Scroll up/down the available devices and choose EP2C35F672C6. Leave other options as they are. Click **Finish**.

Create a Half Adder

1. File → New. New file option window appears.
2. Select Block Diagram/ Schematic File. Click OK. BDF Editor window appears.
3. Double click anywhere on the window. Select Symbol window appears.
4. Type xor in the Name box. Click Ok. Drag the XOR symbol and put it the middle of the window.
5. Repeat steps 3 and 4 but with and2 symbol name to put a 2-input AND symbol at the bottom of the XOR symbol.
6. Input ports and output ports are symbols with names "input" and "output", respectively. Thus, repeat steps 3 and 4 with symbol name "input" and "output" to put two input ports and two output ports on the window. Click and drag the symbols on the window to match the BDF diagram shown below.



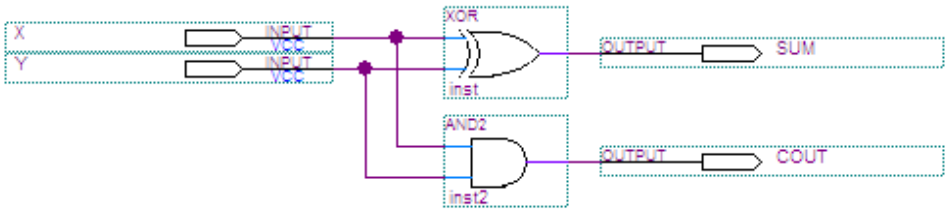
7. Click the orthogonal node tool in the list of tool on the left of the BDF window (see above). To connect point A to point B, position the cursor on point A, click and hold the left button of the mouse, drag the line and drop it at point B.

   Example: To connect input symbol to the XOR symbol input, position the cursor on the right end point of the input symbol, click and hold the left mouse button, drag the line and drop it on the input of the XOR symbol.

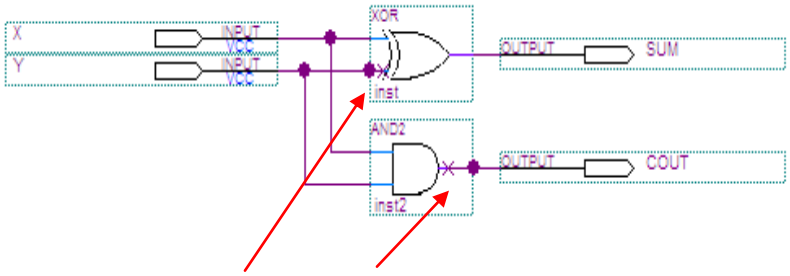
   Follow the steps above connect the input symbols and the output symbols to the XOR and AND symbols.

8. Click the cursor in the list of tool on the left of the BDF window (see above). Double click the input symbol and change the input name to X. Double click the other input symbol and change the input name to Y. Repeat the steps to name the output names to SUM and COUT, respectively.

   Now, your bdf file must look similar to the one shown below:



   To help you in identifying erroneous line connection, an example is shown below:



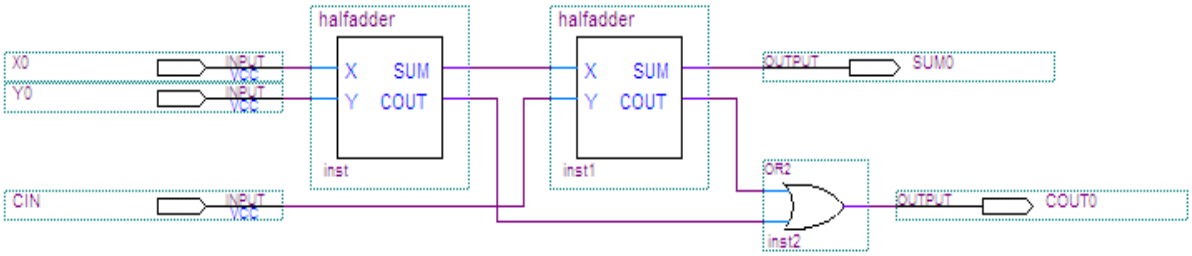
   The errors could be identified as “x” on the lines. If you see these “x”s, delete the lines and redraw them.

9. Save the bdf file as halfadder.bdf.

10. File → Create/Update → Create Symbol Files for Current File. Create symbol file window appears. Note that the filename shown is “halfadder”. Click Save. An acknowledgement that the halfadder symbol has been created appears. Click Ok.

Create a 2-bit adder using two halfadders and one OR gates.

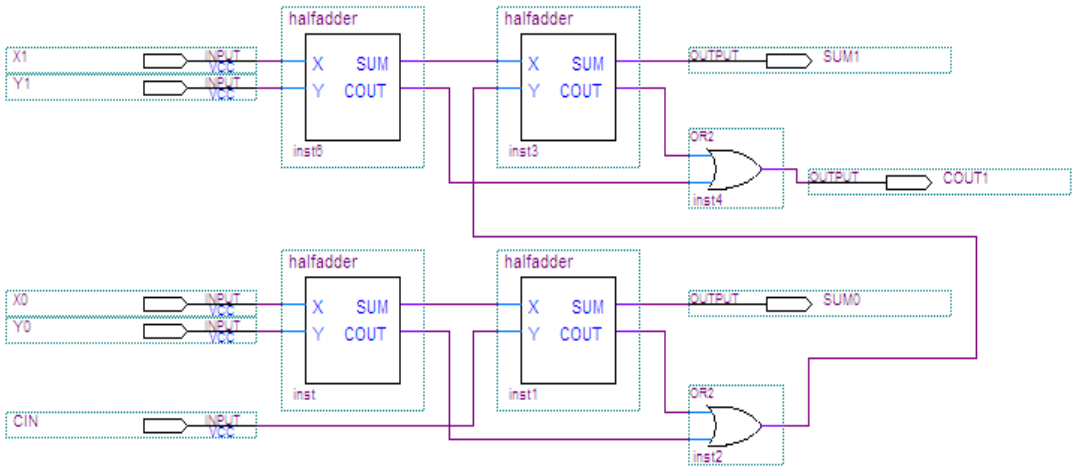
1. File → New. New file option window appears.
2. Select Block Diagram/ Schematic File. Click OK. BDF Editor window appears.
3. Double click anywhere on the window. Select Symbol window appears.
4. Expand the Project folder in Libraries window, click halfadder, and click Ok. Put the halfadder symbol in the middle of the bdf window.
5. Repeat steps 3 and 4 to put another halfadder symbol on the bdf window.
6. Put a 2-input OR symbol (OR2) on the bdf window. You should know how to do this by now. If not, review the previous steps on how to put an AND symbol on the bdf window.
7. Put three input symbols and two output symbols on the bdf window and connect them to the respective inputs and outputs. Rename the input symbols and output symbols as shown on the bdf diagram below:



Your fulladder bdf file must look similar to the one shown above. If there is a mismatch, go back to the previous steps to fix the errors.

8. You could have followed steps 3 to 7 to create another one-bit fulladder. But you would save time by using the copy and paste facilities. Select the entire fulladder you have created earlier by putting the cursor on the left hand upper corner, left click and drag the selection window to cover all symbols in the fulladder. Edit → Copy.
9. Move the cursor to a position above and far from the fulladder and paste the selection. Position this new fulladder (bit 1) so that it lines up with the fulladder (bit 0) below.
10. Delete Cout0 symbol for fulladder (bit 0) and delete CIN symbol for fulladder (bit1) but leave the lines intact. Connect the lines left open after deleting Cout0 and CIN.

Your 2-bit adder bdf file must look similar to the one shown below. If there is a mismatch, go back to the previous steps to fix the errors.



11. Save your 2-bit adder file and name it Tutorial02A.bdf. Note that the top design name must be the same as the project name. Here, the top design is a 2-bit adder and our project name is Tutorial02A.

Compile the 2-bit adder

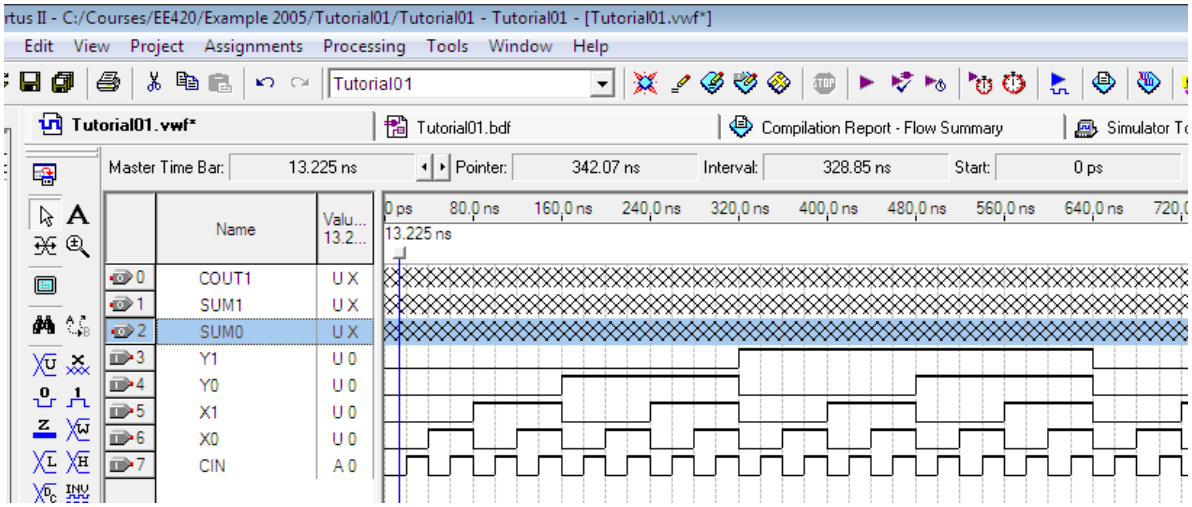
Processing → Start Compilation. If there is errors in the message window, fix the errors and compile the design again. Ignore the warnings.

Simulate the 2-bit adder (Timing Simulation)

1. File → New. New file option window appears.
2. Click Other Files tab and select Waveform Vector File. Click Ok. Waveform Vector File (wvf) windows appears.
3. Left click anywhere in the Name pane (left of vwf window) and click Insert → Insert Node or Bus. Insert Node or Bus window appears.
4. Click Node Finder. Node Finder window appears.
5. Select Pins: all in the Filter selection box and click List. Move all signals from the left pane (Nodes Found) to the right pane (Nodes Selected) by clicking >>. Click OK. On the Insert Node or Bus window click OK again. The vwf window appears.

6. Arrange the signals in the Name pane in logical order: COUT1, SUM1, SUM0, Y1, Y0, X1, X0, CIN where COUT1 is at the top. To move the signal up or down, select the signal, click and drag it up or down.
7. Right click on CIN and Value → Count Value. Count Value window appears. Click Timing tab and change Count Every value to 20 ns. Click OK.
8. Repeat step 6 for X0 but change the Multiplied by value to 2.
9. Repeat step 6 for X1 but change the Multiplied by value to 4.
10. Repeat step 6 for Y0 but change the Multiplied by value to 8.
11. Repeat step 6 for Y1 but change the Multiplied by value to 16.
12. View → Zoom Out as many times as necessary so that you could see all possible combinations of Cin through Y1 in the interval 0 to 640 ns.

Your Tutorial02A vwf file must look similar to the one shown below. If there is a mismatch, go back to the previous steps to fix the errors.



13. Save your Tutorial02A.vwf file.
14. First, you will do the timing simulation where you can see the signal delay from inputs to outputs (Simulation mode: Timing). Processing → Simulator Tool. Simulator Tool window appears. Check the Overwrite simulation input file with simulation results box. Click Start. If you do not see the Start button, close the message window at the bottom.
15. If the simulation is successful, click Open to open the vwf file and see the result of simulation. If you do not see the Open button, close the message window at the bottom.
16. Check the result of simulation and take notes of the delay.

Simulate the 2-bit adder (Functional Simulation)

1. Now, you will do the functional simulation where all output signals change (if any) at the same time as the input signals. Processing → Simulator Tool. Simulator Tool window appears. Select Functional in the Simulation mode list. Click Generate Functional Simulation Netlist. When the compilation is completed, click Start. If you do not see the Start button, close the message window at the bottom. Note that the compilation should not have any errors because you have compiled the design successfully for the timing simulation earlier.
2. If the simulation is successful, click Open to open the vwf file and see the result of simulation. If you do not see the Open button, close the message window at the bottom.
3. Check the result of simulation and take notes of the zero-time delay.

Pins Assignments for DE2 Board

1. Assignments → PINs (Click PINs under Assignments)
2. Double click the blank box under Location for each of the pin names in the Table below and select the respective pin numbers. For example, to set pin number for CIN. Double click the blank box under Location for CIN node name and select PIN_N25.
3. Click OK.

Pin Assignment Table

DE2 Board Component Names	Tutorial02A Signal Names	Pin Numbers
SW[4]	Y1	PIN_AF14
SW[3]	Y0	PIN_AE14
SW[2]	X1	PIN_P25
SW[1]	X0	PIN_N26
SW[0]	CIN	PIN_N25
LEDR[2]	COUT1	PIN_AB21
LEDR[1]	SUM1	PIN_AF23
LEDR[0]]	SUM0	PIN_AE23

Fit the Project to FPGA

To launch Fitter, click Processing → Start → Start Fitter.

Assemble Project

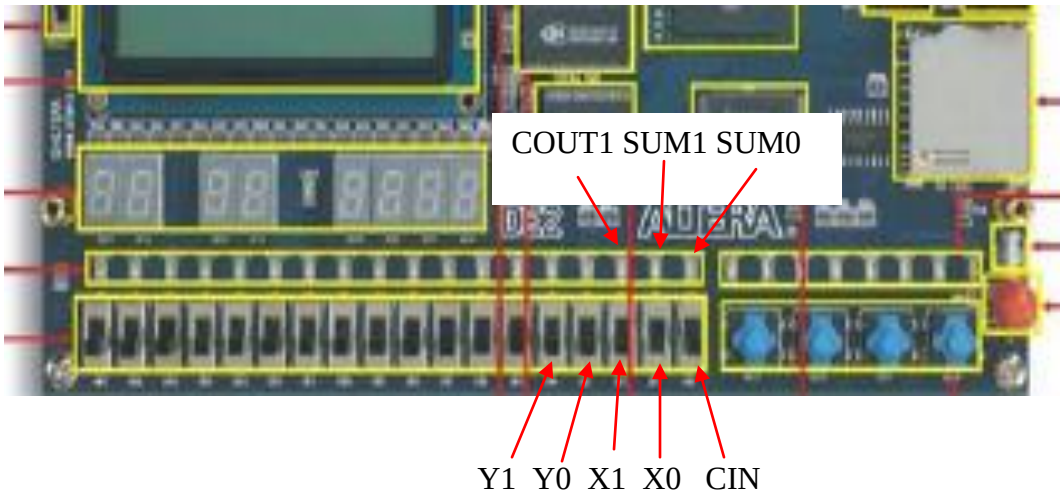
Assembler will generate the programming files (files used to configure the FPGA. For Cyclone II, it is the .sof file)

To launch Assembler, click Processing → Start → Start Assembler.

Downloading a Hardware Image to the FPGA Cyclone II 2C35F672C6

To download the Tutorial02A.sof File to the DE2 board, follow these steps:

1. On the Tools menu, click **Programmer**.
2. Click **Hardware Setup**.
3. Select **USB-Blaster** in the **Currently selected hardware** list and click **Close**.
4. Turn on (check) **Program/Configure,** located on the same row as Tutorial02A**.sof**.
5. Click **Start**.
6. Upon download success, the Messages window will display "Info: successfully performed operation(s)." If you do not see this message, check your cable connections and the Quartus II Programmer hardware setup.
7. Close the **Programmer**.

Verify 2-bit adder (Hardware verification)

Close and open SW3, SW2, SW1, and SW0 to give 16 possible combinations. For each combination, check the LEDR2, LEDR1, and LEDR0 if they match the truth table for a 2-bit adder.